

ANX-PR/CL/001-01

GUÍA DE APRENDIZAJE

ASIGNATURA

83000087 - Hydrodynamics Of Hulls And Propellers (split)

PLAN DE ESTUDIOS

08IN - Master Universitario En Ingenieria Naval Y Oceanica

CURSO ACADÉMICO Y SEMESTRE

2025/26 - Segundo semestre

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1. Datos descriptivos

1.1. Datos de la asignatura

Nombre de la asignatura	83000087 - Hydrodynamics Of Hulls And Propellers (Split)
No de créditos	4 ECTS
Carácter	Obligatoria
Curso	Primer curso
Semestre	Segundo semestre
Período de impartición	Febrero-Junio
Idioma de impartición	Inglés/Castellano
Titulación	08IN - Master Universitario en Ingeniería Naval y Oceanica
Centro responsable de la titulación	08 - E.T.S. De Ingenieros Navales
Curso académico	2025-26

2. Profesorado

2.1. Profesorado implicado en la docencia

Nombre	Despacho	Correo electrónico	Horario de tutorías *
Antonio Souto Iglesias (Coordinador/a)	El habitual	antonio.souto@upm.es	Sin horario. Ver horario general de tutorías.
Javier Calderon Sanchez	El habitual	javier.calderon@upm.es	Sin horario. Ver horario general de tutorías.

* Las horas de tutoría son orientativas y pueden sufrir modificaciones. Se deberá confirmar los horarios de tutorías con el profesorado.

2.2. Personal investigador en formación o similar

Nombre	Correo electrónico	Profesor responsable
Portillo Juan, Adrian	adrian.portillo.juan@upm.es	Souto Iglesias, Antonio

3. Competencias y resultados de aprendizaje

3.1. Competencias

CG1 - Que los estudiantes sepan aplicar los conocimientos adquiridos y su capacidad de resolución de problemas en entornos nuevos o poco conocidos dentro de contextos más amplios (o multidisciplinares) relacionados con su área de estudio. Students should be able to apply their acquired knowledge and problem-solving skills in new or unfamiliar environments within broader (or multidisciplinary) contexts related to their area of study.

CG2 - Que los estudiantes sean capaces de integrar conocimientos y enfrentarse a la complejidad de formular juicios a partir de una información que, siendo incompleta o limitada, incluya reflexiones sobre las responsabilidades sociales y éticas vinculadas a la aplicación de sus conocimientos y juicios. That students are able to integrate knowledge and face the complexity of making judgments based on incomplete or limited information, including reflections on the social and ethical responsibilities linked to the application of their knowledge and judgments.

CG3 - Que los estudiantes sepan comunicar sus conclusiones- y los conocimientos y razones últimas que las sustentan a públicos especializados y no especializados de un modo claro y sin ambigüedades. That students know how to communicate their conclusions - and the ultimate knowledge and reasons that support them - to specialized and non-specialized audiences in a clear and unambiguous manner.

CG4 - (S1) Que los estudiantes posean las habilidades de aprendizaje que les permitan continuar estudiando de un modo que habrá de ser en gran medida autodirigido o autónomo.

CTUPM01 - (S2) Creatividad. Los estudiantes deben resolver de forma nueva, original y aportando valor, situaciones o problemas en el ámbito de la ingeniería. Creativity. Students must solve situations or problems in the field of engineering in a new, original and valuable way.

CTUPM02 - (S3) Organización y planificación. Los estudiantes fijan objetivos, con la planificación y programación de actividades (tiempo y fases) y con la organización y gestión de los recursos necesarios para alcanzarlos. Organization and planning. Students set objectives, with the planning and scheduling of activities (time and phases) and with the organization and management of the resources needed to achieve them.

CTUPM03 - (S4) Liderazgo. Los estudiantes dirigen y coordinan personas para que trabajen con entusiasmo en la consecución de objetivos en pro del bien común. Leadership. Students direct and coordinate people to work enthusiastically to achieve goals for the common good.

CTUPM04 - (S5) Uso de la lengua inglesa. Los estudiantes establecen conversaciones con nativos sin tener problemas de comunicación adicionales tanto de forma oral como escrita. Use of the English language. Students establish conversations with native speakers without having additional communication problems both orally and in writing.

CTUPM05 - (S6) Uso de las tecnologías de la información y comunicación (TIC). Los estudiantes aplican conocimientos tecnológicos necesarios de manera que les permitan desenvolverse cómodamente y afrontar los retos que la sociedad les va a imponer en su quehacer profesional empleando la informática. Use of information and communication technologies (ICT). Students apply the necessary technological knowledge in a way that allows them to perform comfortably and meet the challenges that society will impose on them in their professional work using information technology.

CTUPM06 - (S7) Comunicación oral y escrita. Los estudiantes transmiten conocimientos y expresan ideas y argumentos de manera clara, rigurosa y convincente, tanto de forma oral como escrita, utilizando los recursos gráficos y los medios necesarios adecuadamente y adaptándose a las características de la situación y de la audiencia. Oral and written communication. Students transmit knowledge and express ideas and arguments in a clear, rigorous and convincing manner, both orally and in writing, using the necessary graphic resources and media appropriately and adapting to the characteristics of the situation and the audience.

CTUPM08 - Trabajo en equipo. Los estudiantes desarrollan la capacidad para trabajar en equipo, integrarse y colaborar de forma activa en la consecución de objetivos comunes. Teamwork. Students develop the ability to work in teams, integrate and collaborate actively in the achievement of common objectives.

CTUPM09 - Resolución de problemas. Los estudiantes son capaces de identificar o proponer un problema, y tienen el conocimiento sobre diferentes alternativas metodológicas y estratégicas para resolverlo. Problem solving. Students are able to identify or propose a problem, and have the knowledge about different methodological and strategic alternatives to solve it.

CTUPM13 - Trabajo en contextos internacionales. Los estudiantes son capaces de integrarse en un grupo o equipo, colaborando y cooperando con otros. Tienen la capacidad para trabajar con estudiantes de otras disciplinas y de aceptar la diversidad social y cultural. Work in international contexts. Students are able to integrate

into a group or team, collaborating and cooperating with others. They have the ability to work with students from other disciplines and to accept social and cultural diversity.

3.2. Resultados del aprendizaje

RA10 - COMPETENCIAS: Capacidad para revisar y aplicar a la resistencia de un buque los conceptos de placa plana laminar y turbulenta

RA11 - COMPETENCIAS: Capacidad para identificar, calcular y analizar la resistencia viscosa y el efecto de forma

RA15 - COMPETENCIAS: Capacidad para identificar, calcular y analizar el efecto de los apéndices así como su extrapolación

RA16 - HABILIDADES Y DESTREZAS: Saber manejar software de métodos semiempíricos de estimación de resistencia

RA17 - HABILIDADES Y DESTREZAS: Saber manejar dicho software para optimización de dimensiones principales de las carenas

RA18 - HABILIDADES Y DESTREZAS: Saber manejar software de diseño y optimización de propulsores mediante series sistemáticas

RA14 - COMPETENCIAS: Capacidad para identificar, calcular y analizar las diferentes componentes de la resistencia al avance de un buque

4. Descripción de la asignatura y temario

4.1. Descripción de la asignatura

The course consists of two main parts: Resistance and Propulsion.

The first part deals with the different elements that make up the resistance of a ship at a certain speed, which would allow the ship to obtain its towing power.

The second part focuses on the design of the propulsion propeller that would allow the ship to move forward at that speed. The classes will take place in the classroom and in the computing centre.

As a final part of the course, there will be a lesson dedicated to shallow depth effects: squat and sea trial effects.

The course also has practical sessions in the ETSI Naval Towing tank, related to the contents of the course.

In addition, students will be offered the possibility of doing an internship in the Cavitation Tunnel of "Canal de Experiencias Hidrodinámicas de El Pardo".

Also, this subject will be worked together with the Advanced Hydrodynamics of the Vessel in terms of team work. It should be noted that both courses are the split of the higher-level course ?Ship Hydrodynamics? in this study plan.

4.2. Temario de la asignatura

1. Hydrodynamic forces: shear and normal stresses
 - 1.1. Resistance from stresses integral. Form Drag.
 - 1.2. Ship resistance and its components. Order of magnitude of Friction drag
 - 1.3. Dimensional analysis
 - 1.4. Methodologies to estimate ship resistance.
2. Viscous resistance
 - 2.1. Boundary layer thickness
 - 2.2. Flat plate frictional resistance.
 - 2.2.1. Laminar solution. Blasius
 - 2.2.2. Turbulent solution. Friction lines. ITTC 57
 - 2.3. Separation or detachment of the boundary layer
3. Wave resistance
 - 3.1. Scaling of wave resistance.
 - 3.2. Manual estimation of total resistance
 - 3.3. Kelvin wave system
 - 3.4. Interference effects. Wigley experiments
4. Model-vessel correlation methods
 - 4.1. Froude correlation method
 - 4.2. Hughes correlation method

- 4.3. Methods for the experimental determination of the form factor
- 4.4. ITTC-78 Recommendations
- 4.5. Semi-empirical methods for resistance estimation: Holtrop and others
- 4.6. Systematic Series for Hull Design
- 5. Influence of vessel dimensions and shapes on resistance
 - 5.1. Main dimensions and dimensionless relationships
 - 5.2. Influence of the sectional area curve
 - 5.3. Influence of the longitudinal position of the center of buoyancy
 - 5.4. Influence of waterplane
- 6. Bulbous bow. Transversal sections
 - 6.1. Bulbous bow. Functional considerations.
 - 6.2. Influence of hull shape
- 7. Power types and propeller geometry
 - 7.1. Propeller fundamentals as a propulsive element: thrust and torque
 - 7.2. Power types and efficiencies: general description
 - 7.3. Helical surfaces
 - 7.4. Graphical representation of the propeller
 - 7.5. Geometrical relations
 - 7.6. Velocity and force diagram in a radial section of a propeller.
- 8. Propeller in open water conditions
 - 8.1. Dimensionless parameters. K_T , K_Q diagrams
 - 8.2. Influence of the Reynolds number
- 9. Hull-propeller interaction
 - 9.1. Wake Fraction
 - 9.2. Thrust deduction fraction
 - 9.3. Relative rotative efficiency
 - 9.4. Quasi-propulsive Efficiency
 - 9.5. Self-propulsion test.
 - 9.5.1. Friction deduction correction.

9.5.2. Effective Wake. Thrust identity

10. Cavitation

10.1. Definition

10.2. Local cavitation number

10.3. Influence of expanded area ratio on cavitation

10.4. Cavitation criteria: Keller & Burrill.

11. Systematic propeller series

11.1. Wageningen B-screw series

11.2. Gawn series

11.3. Gawn-Burrill series

11.4. Newton-Rader series

12. Manual design of a propeller using systematic series.

12.1. Design scenarios

12.1.1. Theoretical Limit to propeller efficiency based on axial momentum theory.

12.2. Hull-engine-propeller coupling in manual calculation / Motor curves / available torque check

13. Off-design conditions.

13.1. Bollard Pull and Trawling Conditions.

13.2. RPM reduction

13.3. Increase of resistance due to fouling.

14. Shallow-water conditions and ship resistance.

14.1. Squat / Barrass method

14.2. Sea trials / Lackenby and Schlichting Methods.

15. Nozzled Propellers

15.1. Fundamentals: Nozzle forces.

15.2. Wageningen Nozzled propeller series

5. Cronograma

5.1. Cronograma de la asignatura *

Sem	Actividad tipo 1	Actividad tipo 2	Tele-enseñanza	Actividades de evaluación
1	Course presentation Duración: 01:00 LM: Actividad del tipo Lección Magistral Lesson 1 Duración: 02:00 LM: Actividad del tipo Lección Magistral			
2	Lessons 1 & 2 Duración: 02:00 LM: Actividad del tipo Lección Magistral Exercises. Duración: 01:00 PR: Actividad del tipo Clase de Problemas			
3	Lesson 2. Duración: 01:00 LM: Actividad del tipo Lección Magistral	Software to estimate resistance. Duración: 02:00 PL: Actividad del tipo Prácticas de Laboratorio		Associated Moodle Task. Resistance Software TI: Técnica del tipo Trabajo Individual Evaluación Progresiva Presencial Duración: 00:00
4	Lesson 2 Duración: 02:00 LM: Actividad del tipo Lección Magistral Exercises Duración: 01:00 PR: Actividad del tipo Clase de Problemas			
5	Lesson 3 Duración: 01:00 LM: Actividad del tipo Lección Magistral Exercises Duración: 02:00 PR: Actividad del tipo Clase de Problemas			
6	Lesson 4. Duración: 01:00 LM: Actividad del tipo Lección Magistral Exercises Duración: 02:00 PR: Actividad del tipo Clase de Problemas			
7	Lessons 5, 6 Duración: 02:00 LM: Actividad del tipo Lección Magistral Team Work. Part 1. Presentation. Ship resistance estimation for a case study specific for each group. Duración: 01:00			Team Work. Part 1. Presentation. Ship resistance estimation for a case study specific for each group. PG: Técnica del tipo Presentación en Grupo Evaluación Progresiva Presencial Duración: 00:00

	OT: Otras actividades formativas / Evaluación			
8	Visit to CEHIPAR Duración: 02:00 OT: Otras actividades formativas / Evaluación Midterm1. Lessons 1-6 Duración: 01:00 OT: Otras actividades formativas / Evaluación	Resistance test Duración: 01:00 PL: Actividad del tipo Prácticas de Laboratorio		Midterm1. Lessons 1-6 EX: Técnica del tipo Examen Escrito Evaluación Progresiva Presencial Duración: 00:00
9	Lesson 7 Duración: 01:00 LM: Actividad del tipo Lección Magistral Lesson 8 Duración: 01:00 LM: Actividad del tipo Lección Magistral Exercises Duración: 01:00 PR: Actividad del tipo Clase de Problemas	Open water tests. Lab Practice Duración: 01:00 PL: Actividad del tipo Prácticas de Laboratorio		
10	Lesson 9. Duración: 03:00 LM: Actividad del tipo Lección Magistral	Self-Propulsion test. Lab practice. Duración: 01:00 PL: Actividad del tipo Prácticas de Laboratorio		
11	Lesson 10 Duración: 01:00 LM: Actividad del tipo Lección Magistral Lesson 11 Duración: 01:00 LM: Actividad del tipo Lección Magistral Exercises. Duración: 01:00 PR: Actividad del tipo Clase de Problemas			
12	Lesson 12 Duración: 01:00 LM: Actividad del tipo Lección Magistral	Software to design propeller Duración: 02:00 PL: Actividad del tipo Prácticas de Laboratorio		Associated Moodle Task. Propeller design software TI: Técnica del tipo Trabajo Individual Evaluación Progresiva Presencial Duración: 00:00
13	Lesson 13 Duración: 02:00 LM: Actividad del tipo Lección Magistral			
14	Lesson 14 Duración: 02:00 LM: Actividad del tipo Lección Magistral Prácticas y Ejercicios Duración: 01:00 PR: Actividad del tipo Clase de Problemas			
15	Exercises. Duración: 02:00 PR: Actividad del tipo Clase de Problemas Team Work Part 2. Presentation. Propeller design for the same case like Team Work Part 1. Duración: 01:00			Team Work. Part 2. Presentation. Propeller design for Part 1 case study. PG: Técnica del tipo Presentación en Grupo Evaluación Progresiva Presencial Duración: 00:00

	OT: Otras actividades formativas / Evaluación			
16				Lab practices TG: Técnica del tipo Trabajo en Grupo Evaluación Progresiva y Global Presencial Duración: 00:00 Progressive Assessemnt. Total Grade of Theory (non-lab) part Grade obtained with the weights described above, excluding ETSIN lab practices. OT: Otras técnicas evaluativas Evaluación Progresiva Presencial Duración: 00:00
17	Global evaluation test. Midterm 1 Duración: 01:00 OT: Otras actividades formativas / Evaluación Midterm 2: lessons 7- end. This exam is also part of the global assessmen test. Duración: 01:00 OT: Otras actividades formativas / Evaluación			Midterm 2: lessons 7- end. This exam is also part of the global assessmen test. EX: Técnica del tipo Examen Escrito Evaluación Progresiva Presencial Duración: 00:00 Global evaluation test. Midterms 1 & 2 will be taken separately. Their relative weight will be the same EX: Técnica del tipo Examen Escrito Evaluación Global Presencial Duración: 00:00

Para el cálculo de los valores totales, se estima que por cada crédito ECTS el alumno dedicará dependiendo del plan de estudios, entre 26 y 27 horas de trabajo presencial y no presencial.

6. Actividades y criterios de evaluación

6.1. Actividades de evaluación de la asignatura

6.1.1. Evaluación (progresiva)

Sem.	Descripción	Modalidad	Tipo	Duración	Peso en la nota	Nota mínima	Competencias evaluadas
3	Associated Moodle Task. Resistance Software	TI: Técnica del tipo Trabajo Individual	Presencial	00:00	5%	0 / 10	CTUPM04 CTUPM05
7	Team Work. Part 1. Presentation. Ship resistance estimation for a case study specific for each group.	PG: Técnica del tipo Presentación en Grupo	Presencial	00:00	10%	0 / 10	CTUPM04 CTUPM09 CTUPM02 CTUPM03 CTUPM06 CTUPM05 CTUPM13 CTUPM08
8	Midterm1. Lessons 1-6	EX: Técnica del tipo Examen Escrito	Presencial	00:00	22.5%	3 / 10	CG1 CG2 CG3 CG4 CTUPM04 CTUPM05 CTUPM09 CTUPM01
12	Associated Moodle Task. Propeller design software	TI: Técnica del tipo Trabajo Individual	Presencial	00:00	5%	0 / 10	CTUPM04 CTUPM05
15	Team Work. Part 2. Presentation. Propeller design for Part 1 case study.	PG: Técnica del tipo Presentación en Grupo	Presencial	00:00	10%	0 / 10	CTUPM04 CTUPM09 CTUPM02 CTUPM03 CTUPM06 CTUPM05 CTUPM13 CTUPM08
16	Lab practices	TG: Técnica del tipo Trabajo en Grupo	Presencial	00:00	25%	5 / 10	CG1 CTUPM03 CTUPM06 CTUPM05 CTUPM13 CTUPM08 CTUPM04

							CTUPM02
16	Progressive Assessemnt. Total Grade of Theory (non-lab) part Grade obtained with the weights described abve, excluding ETSIN lab practices.	OT: Otras técnicas evaluativas	Presencial	00:00	%	5 / 10	CG1 CG2 CG4 CG3
17	Midterm 2: lessons 7- end. This exam is also part of the global assessmen test.	EX: Técnica del tipo Examen Escrito	Presencial	00:00	22.5%	3 / 10	CG1 CG2 CG3 CG4 CTUPM04 CTUPM05 CTUPM09 CTUPM01

6.1.2. Prueba evaluación global

Sem	Descripción	Modalidad	Tipo	Duración	Peso en la nota	Nota mínima	Competencias evaluadas
16	Lab practices	TG: Técnica del tipo Trabajo en Grupo	Presencial	00:00	25%	5 / 10	CG1 CTUPM03 CTUPM06 CTUPM05 CTUPM13 CTUPM08 CTUPM04 CTUPM02
17	Global evaluation test. Midterms 1 & 2 will be taken separately. Their relative weight will be the same	EX: Técnica del tipo Examen Escrito	Presencial	00:00	75%	5 / 10	CG1 CG2 CG3 CG4 CTUPM02 CTUPM03 CTUPM04 CTUPM05 CTUPM06 CTUPM09 CTUPM01

6.1.3. Evaluación convocatoria extraordinaria

Descripción	Modalidad	Tipo	Duración	Peso en la nota	Nota mínima	Competencias evaluadas
Exam + Lab Practice. Assessment method is a repetition of Global Examination (Ordinary call) with same minimum grades for theory and lab practices (both need be passed as in the Global Examination - Ordinary Call)	OT: Otras técnicas evaluativas	Presencial	02:00	100%	5 / 10	CTUPM09 CTUPM02 CG1 CTUPM04 CTUPM01 CG2 CG4 CG3 CTUPM03 CTUPM06 CTUPM05 CTUPM13 CTUPM08

6.2. Criterios de evaluación

Visit to CEHIPAR and subsequent report provide up to 0.5p on top the grade.

A rubric is provided for assessing group works:

Moodle thread: number of entries, quality, periodicity, documenting the context of the choice, references, etc., uploading files before the presentation 30%.

Defense, assessing the technical aspects of the work, its depth, the presentation file, quality of the defence, ability to answer the questions (all members of the group can answer them), etc... 70%

Regarding Moodle, a single thread will be maintained for both DB (Ship Dynamics) and HAB (Advanced Ship

Hydrodynamics) teamworks. It will be in HAB Moodle space.

If the Head of Studies (at the request of a student) indicates that an absence from a practical is justified, the following measures will be taken:

- 1 First of all, this student will be formally withdrawn from the corresponding group.
- 2 A video will be shown to the person in question, some information will be given to him/her, and a short script of the practical will be given to him/her.
3. The person in question will have to make the corresponding analysis and practice report.
4. They will also be given an article from a scientific journal that has a connection with the practice and will be asked to summarise it in 600-800 words, highlighting the connections between the article and the topic of the internship.

If the Head of Studies considers that the absence is not justified in the ordinary exam session, the student will fail the practice for not attending, with all that this implies with respect to the ordinary exam session.

All laboratory practices must be passed.

In the global evaluation scheme, the scenario of considering the mid-term exams taken in the global evaluation as the equivalent items of the progressive evaluation will be simulated with all that this implies.

The maximum of the two scenarios (global and progressive) will be taken as the final grade.

The midterms taken during the course are kept for the progressive evaluation scenario of the ordinary global and the extraordinary evaluation in case the student does not attend any of those midterms.

If he/she does show up, what counts is the grade obtained in that midterm in the global test for both scenarios.

For the global evaluation scenario, if the student does not take a midterm, his/her grade is zero.

The grade corresponding to Midterm 1 taken in the Ordinary Call for "Global Assessment Test" will NOT be kept for the Extraordinary Examination.

For the submission and defense of group works, several submission windows will be opened in coordination with the students' various duties.

The Extraordinary Examination functions as an exact repetition of the Global Assessment Test in the Ordinary Examination.

The only exception is that the student may wish not to take the second midterm, in case he/she wants to keep that grade.

If one of the parts ("laboratory practices" and "the rest") is failed, the student will get the grade corresponding to the failed part. Therefore, the maximum grade in the global and progressive evaluation if one part is failed is the minimum of both parts.

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For assignments, a Moodle task will be enabled for uploading evidence of completion. Failure to do so within the established deadline cannot be remedied.

For the submission and defense of group works, several submission windows will be opened.

If one of the parts ("laboratory practicals" and "the rest") is failed, the student will get the mark corresponding to the failed part. Therefore, the maximum mark in the global and progressive evaluation if one part is failed is the minimum of both parts.

The practices of resistance test, open-water propeller and self-propulsion will be carried out in ETSIN towing tank, by groups of students.

Prior to the realization of the practices, the students will carry out a "questionnaire" on the contents of the practice. Afterwards, the students will write the corresponding reports which, once evaluated, will contribute to the corresponding percentage of the final grade of the course.

Regarding the practical grade:

50% = Reports: (trailer + isolated propellant + 2*car)/4

50% = average of the pre-practice tests (V/F tests).

All three practices must be attended and the reports must be approved.

The towing and open water reports are in group.

The self-propulsion report is individual.

If the lab grade is a pass, such grade is KEPT FROM ONE academic year TO ANOTHER IF THE COURSE IS FAILED.

7. Recursos didácticos

7.1. Recursos didácticos de la asignatura

Nombre	Tipo	Observaciones
Apuntes de la Asignatura de D. Antonio Baquero Mayor	Bibliografía	
Principles of Naval Architecture. SNAME	Bibliografía	
Manual de Prácticas de Laboratorio de Luis Perez Rojas	Bibliografía	

Ship Resistance and Propulsion de A. Molland	Bibliografía	
Marine Propellers and Propulsion, de J. Carlton	Bibliografía	
Presentations used during the Lessons.	Bibliografía	
Holtrop method	Bibliografía	Holtrop, J., 1992, Hydrodynamics in ship design : training course. Wageningen Holanda Marin (ejemplar en biblioteca ETSIN) (documento interesante en general, y con una parte sobre métodos semi-empíricos).
Maxsurf Resistance	Otros	Software
FreeCAD	Otros	Software
JSDN	Otros	Propeller Software

8. Otra información

8.1. Otra información sobre la asignatura

The timetable follows a theoretical planning of the subject that may undergo modifications during the course.

Attendance sheets will be signed for the laboratory sessions, as well as for the different evaluable tasks.

The software sessions may take place at any class time or in the common time. They will be announced in advance.

For the team works, prior to the defence, the required files (presentation, software, etc.) will be handed in.

Goals and targets (of the 2030 Agenda for Sustainable Development)

Goal 2. End hunger, achieve food security and improved nutrition and promote sustainable agriculture

2.3 By 2030, double the agricultural productivity and incomes of small-scale food producers, particularly women, indigenous peoples, family farmers, livestock keepers and fishers, including through secure and equitable access to land, other productive resources and inputs, and knowledge, financial services, markets and opportunities for value addition and off-farm employment.

The course prepares students for better fishing vessel designs, which has an impact on this goal. In fact, the course coordinator has participated in 2023 in a UN-FAO project to improve the energy efficiency of artisanal fishing vessels for developing countries.

Goal 4. To ensure inclusive and equitable quality education and promote lifelong learning opportunities for all.

4.3 By 2030, ensure equal access for all men and women to quality technical, vocational and higher education, including university education

This objective is being pursued by seeking to train men and women equally in the technical training associated with the subject.

Goal 5. Achieve gender equality and empower all women and girls

5.1 End all forms of discrimination against all women and girls everywhere.

This objective is pursued by seeking to train men and women equally in the technical training associated with the subject.

5.5 Ensure women's full and effective participation and equal opportunities for leadership at all levels of decision-making in political, economic, and public life.

The participation of women in presentations as well as other activities, such as debates, with a public exposure component, will be promoted.

Goal 8. Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all.

8.9 By 2030, develop and implement policies to promote sustainable tourism that creates jobs and promotes local culture and products.

The nautical sector is of great importance for tourism in Spain. The challenges of the subject have an impact on the design of more sustainable recreational boats for tourist use.

Goal 14. Conserve and sustainably use the oceans, seas and marine resources for sustainable development.

14.b Facilitate artisanal fishermen's access to marine resources and markets.

The course prepares students for better fishing vessel designs, which has an impact on this goal. In fact, the course coordinator has participated in 2023 in a UN-FAO project to improve the energy efficiency of artisanal fishing vessels for developing countries.